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3D Product Display's Impact on E-Commerce Websites

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Abstract: On e-commerce websites, the usage of 3D product presentation systems has become a web design trend. At the same time, because the COVID-19 outbreak has limited the number of people going out globally, some firms are forced to host online events such as product debuts, which include the use of 3D features. It is certain that 3D product presentation will be the future development trend for e-commerce websites. Based on a theoretical framework and literature analysis, the study employs a mix of qualitative and quantitative research approaches. Through more creative operations and visual experiences, 3D product display may provide users with a more natural and realistic online product experience. 3D elements enhance product details and make online products more three-dimensional.

The study also discovered that, when compared to traditional two-dimensional pictures for product display, 3D displays can more accurately reflect the characteristics of products in e-commerce and show product details, allowing users to obtain relevant product information in detail and efficiently, thereby improving users' purchase intentions for the same product.

As a result, this research asserts that 3D models for displaying items will continue to be a common trend in the future growth of e-commerce web design. To make 3D product presentation more familiar and efficient, e-commerce merchants must invest more time, money, and human resources in this industry; site designers should consider how to employ 3D technology to create a more immersive web surfing environment for customers.

Keywords - 3D product display, web design, online product experience

I. INTRODUCTION

Before the creation of computers, 3D vision was created. To create three-dimensional figures, mathematics and geometry can correctly track distance and position. The creation of 3D visuals is based on mathematical concepts. The geometry of triangles, which are built on straight lines and composed of points, lies at the heart of 3D. It's worth noting that the more money invested in 3D computer graphics, the more realistic it will be, the greater the credibility, and the better the user experience will be. Furthermore, according to Jon (2013) research, computer power has expanded dramatically, democratizing computer graphics and laying the groundwork for 3D graphic design and exploration. Time, focus, interest, and feeling of experience in website design are highlighted as crucial predictors as the user's, and the visual qualities of the website will affect the emotional response of users, according to Lucian, Anna, Mary, and Victor (2015).

Nike is an example of an online solution that incorporates 3D visuals. Nike values user experience and tailored marketing so much that they examine big data and consumer information to conduct customized marketing to various target groups. On its purchasing website, the company uses computer 3D technology to create 3D visuals of sports shoes, allowing users to mix and match colors to achieve personalization. 3D visualization graphics are employed in various sales platforms, such as the online shopping interface, to make the sales model more personalized by allowing users to try out the 3D positioning and matching of furniture in the area on their own.

One of the most crucial elements to consider when purchasing furniture is the amount of space available. Although 2D product photos can provide the majority of the product information that clients require, the arrangement and color of furniture in the room is far too important to rely on the imagination of customers. These issues are well-solved by 3D product displays. Furthermore, the product's material and texture can be clearly exhibited using the 3D model.

Consumers can select a product from the catalog and place it in any location in the virtual area, adjusting its placement in 360 degrees. Consumers can only interact in one virtual area owing to technical limitations, and they cannot switch between spaces. Users can flexibly change the viewing angle and interact with it by using the right mouse button or screen sliding. The benefit is that the user can see products and product details in 360 degrees with clarity and accuracy. The limits of the 3D product display model in the development were highlighted by Potenziani, Dellepiane, Callieri, and Scopigno (2018). Many evaluators are required due to the vast

range of target groups and the complexity and distinctiveness of the 3D user interface. This viewpoint is shared by Xing, Bo, Feng, and Matteo (2019), who suggest that utilizing controlled experiments to replace audience perceptions can quantify possible benefits.

II. PROBLEM STATEMENT

We are all familiar with the flat display method of two-dimensional photos in e-commerce, but it has the disadvantage of providing a poor interactive experience (Yang & Lee, 2008). According to Klein (2003), incorporating 3D technology into e-commerce product displays may fully exploit the effects that the original e-commerce display cannot, and can be viewed via computers and mobile phones as carriers interact with the presentation. Embrace the true spirit of e-commerce. Users engage with practically all senses and 3D product visualization in the display mode, which presents e-commerce products more comprehensively (Klein, 2003). According to Guttentag (2010), the commercial use of 3D in marketing applications is driven by the ability to deliver more product information. It is also believed that multi-sensory experience can increase consumers' confidence in product selection (Hoch and Deighton 1989).

As the volume of information on the Internet has grown fast, so have the demands for better user engagement and deeper information processing. With dynamic web technologies and network databases, simple static pages to display product information can no longer match this requirement (Zuo & Zhao, 2011). With the advancement of technology and the expansion of application capabilities, a new manner of showing enterprise product information is progressively evolving, with dynamic data updates and three-dimensional interactive operations as key aspects. According to Kang, Shin, and Ponto's (2020) research, the display of products using 3D elements has improved the performance of the website and resulted in significant commercial profits for the relevant company.

The abrupt changes brought forth by covid-19 have thrown the entire physical retail economy into disarray. Brands can no longer wait for customers to come to them; instead, they must push their digitalization at a breakneck speed. Many of the world's most prestigious fashion houses were obliged to adapt and began presenting 3D fashion presentations based on their shopping websites (Lee, 2020). However, 3D modeling is a time-consuming and expensive procedure that necessitates a lot of energy (Corvaglia, 2004). The created 3D model for shopping websites may or may not match the seller's standards, especially if the website has a large number of different types of products (Visinescu, Sidorova, Jones, & Prybutok, 2015). It is not reasonable to presume the production expense is absorbed by the shopping website until the 3D model on the homepage has an actual impact on the customer's buying behavior.

In conclusion, 3D product display compensates for the inadequacies of the 2D image plane display mode's poor interactive experience by providing consumers with a more sensual experience. Product presentation based on 3D models has a lot of potential in the commercial world, but it still has a lot of issues. Solving these issues would improve the visual experience of users, the economic benefits of the website, and even the entire web design profession.

As a result, the research object in this work is the online experience of 3D items in shopping websites.

- Discusses the online experience that customers get from 3D components on shopping websites.
- Summarize and define the influence of 3D elements in web design on the user experience of online products.
- Provide web designers with a mechanism for displaying products on e-commerce sites using 3D elements.
- How does a 3D product display in e-commerce differ from 2D displays like a photo or illustration in terms of user experience?
- How might a 3D product display in e-commerce help to stimulate customers' desire to buy the product (by improving the online product experience)?

III. METHOD AND IMPLEMENTATION

This study is referred to as "multi-method quantitative research" since it collected data using a combination of quantitative approaches, including questionnaires. Meetoo and Temple (2003) defined "multi-method" as "approaches that combine more than just data gathering techniques and are also related with analytical methods" (that is, using the form of interviews). This research technique, according to Johnson and Onwuegbuzie (2004), has a larger, more inclusive, and paradigm nature. As a result, the author adopted a multi-method quantitative study to make the research more comprehensive. In addition, because it integrates responses from a larger sample size, the quantitative technique gives more comprehensive and reliable data than the qualitative approach (Creswell, 2003).

3.1 Data collection

Primary and secondary studies are used to acquire data, with primary data using both quantitative and qualitative methodologies.

3.1.1 Quantitative section

The authors will describe the dimensions of our data analysis, research method for descriptive and inferential statistical analysis based on theoretical framework and literature review, explain the rationale for the choice of t-test method, and finally present the research hypotheses in the data analysis section of the quantitative study.

Descriptive and inferential statistical analysis will be performed on the quantitative data collected. IBM SPSS will be used as a statistical software platform. Histograms and pie charts will be used to represent the users' basic information (age, shopping experience, choice and preference of shopping modes) and the comparison of their scores in each dimension under various product presentation ways for descriptive statistics.

The linear analysis will be performed on the acquired data in the inferential statistics phase, and we will use the paired t-test. Because the goal of the study is to see if there is a discernible difference between the two averages, which could be related to particular traits.

3.1.2 Qualitative section

The authors will study the content of the customer interviews in order to uncover commonalities in the data addressing the benefits and experiences of 3D product display in the qualitative section. When compared to quantitative research, qualitative research focuses on the interpretation of meaning. It can also be used for thorough and dynamic descriptive analysis at the micro-level, which can be used to supplement and expand on the quantitative data. As a result, combining quantitative and qualitative data analysis methodologies can aid in providing thorough responses to research issues (Chen, 1991).

IV. RESULTS

In conjunction with the study approach, we summarize and analyze the questionnaire data and interview feedback information. Answers the first and second research questions in a logical order. The specific analysis section and the relationship between quantitative research, qualitative research, and outcomes are presented next.

4.1 Questionnaire Result (Quantitative Method)

The paired sample T-test is primarily used in the analysis approach. The paired T-test is classified into two types statistically: a paired sample T-test and a paired non-parametric test. The main distinction is whether the data being studied has a normal distribution. The paired sample rank-sum test is a general statistical approach for non-normally distributed data.

The normal test method, which is separated into Kolmogorov-Smirnov and Shapiro-Wilk tests, is determined by the sample size.

The Shapiro-Wilk normality test calls for a sample size of 101. As a result, the experimental data is subjected to the following paired T-test analysis.

The paired sample T-test method was employed to evaluate this question, which began with a comparison of 2D and 3D. Because of the limited sample size, the Shapiro-Wilk test was used, and the results are displayed in Figure 1. The appeal of a 3D product display to users is greater than that of a 2D product display.

<i>Tests of Normality</i>						
	<i>koimogorow-smirnov</i>			<i>shapiro-wilk</i>		
	<i>Stalistic</i>	<i>df</i>	<i>Sig</i>	<i>Stalistic</i>	<i>df</i>	<i>Sig</i>
Are the products displayed in 2D more appealing to you when perusing e-commerce sites?	.181	101	<.001	.912	101	<.001
Are the products displayed in 3D more appealing to you when perusing e-commerce sites?	.225	101	<.001	.891	101	<.001

Figure1

Figure 2 depicts the appeal of different dimensions of product display to consumers from the standpoint of visual impact.

<i>Descriptive Statistics</i>				
	<i>N</i>	<i>25th</i>	<i>Percentiles 50th(Media)</i>	<i>75th</i>
Do you think the 2D product display's visual effects can pique your interest in the product?	101	3.0	4.0	4.5
Do you think the 3D product display's visual effects can pique your interest in the product?	101	3.0	4.0	5.0

Figure 2

The user's use of 3D pages to see products is analyzed using a one-sample T-test. The one-sample T-test can be selected directly in SPSS, and the resulting chart is displayed in Figure 3.

One-Sample Test						
					95% Confidence Interval of the Difference	
	t	df	Sig(2tailed)	Media Difference	Lower	Upper
Have you ever seen 3D solutions utilized to display products on an eCommerce website?	1.439	101	.154	.169	-.06	.40
One-Sample Statistics						
	N	Mean	Std. Deviation	Std.Error Mean		
Have you ever seen 3D solutions utilized to display products on an eCommerce website?	101	3.18	1.176	.118		

Figure 3

In the same way, a one-sample T-test is also performed for question "Do you believe that more product information would be able to pique your interest in purchasing?" to explore whether comprehensive product information can more stimulate users' desire to buy is shown in Figure 4.

One-Sample Test						
					95% Confidence Interval of the Difference	
	t	df	Sig(2tailed)	Media Difference	Lower	Upper
Do you believe that more product information would be able to pique your interest in purchasing?	9.572	101	<.001	1.030	.920	1.24
One-Sample Statistics						
	N	Mean	Std. Deviation	Std.Error Mean		
Do you believe that more product information would be able to pique your interest in purchasing?	101	4.03	1.082	.108		

Figure 4

In conclusion, the authors use SPSS to conduct data analysis and confirm that 3D product display is more appealing to consumers, that 3D product display provides more visual stimulation to users than 2D, that 3D product display can display products more comprehensively, allowing consumers to get more information through the screen, and that more comprehensive information can actively stimulate users' desire to purchase, and that consumers prefer the operating experience of 3D product display to that of 2D.

4.2 Interview Result (Qualitative Method)

We have a total of ten interviewees for qualitative research. They are between the ages of 20 and 30 and come from various industries. Clothing accessories, everyday essentials, and pet supplies are among the items available for purchase on the e-commerce portal. They said that existing 3D technology on e-commerce platforms may display product features, allowing consumers to receive more thorough product information through views rather than reading product descriptions while evaluating the influence of 3D product display on their purchasing decisions. Furthermore, the 3D presentation of goods on the website may more realistically restore goods, reducing the negative impacts of variables such as plane photographic composition and color difference on the appearance of goods. The use of 3D product display technology lessens the interviewees' suspicion of e-commerce goods in two ways, making consumers feel more comfortable when purchasing.

Six of the ten interviewees said that the lack of product details in the photographs presented on e-commerce platforms prevents them from getting the information they need, causing them to abandon their purchase. The use of 3D technology to present goods can help customers get a better sense of what they want to buy. It also prevents consumers from being misled by an imbalanced proportion of plane display or a visual inaccuracy, among other things, so that they have more confidence in online purchase.

V. CONCLUSION

Users' need for information engagement and in-depth acquisition has increased as a result of the growth of the Internet, as indicated in the problem statement. When standard 2D product interfaces fail to meet the expectations of users, 3D product display solutions are increasingly adopted. Simultaneously, in the face of Covid-19's influence on physical storefronts, marketers must find more appealing ways to display their products. Many big fashion firms, for example, have established 3D fashion weeks based on their purchasing websites, according to Lee (2020). According to Hochand Deighton (1989), a multi-sensory experience can boost consumer trust in product selections. According to Klein (2003), 3D product visualization may interact with practically all of the user's senses. According to the article's quantitative and qualitative research, 3D product display can provide users with a better operation and visual experience, resulting in a better interactive experience, allowing users to obtain more real and comprehensive product information, and reducing the gap between online and physical products. Increasing product credibility and consumer confidence in product selections, and so encouraging consumer purchases, while also increasing brand loyalty.

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